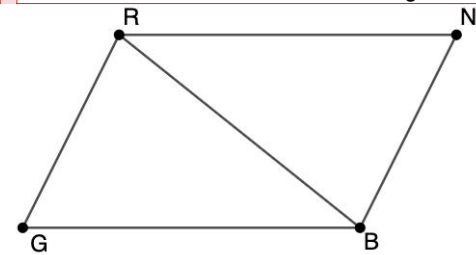


Quadrilateral $RNBG$ is shown with diagonal $\overline{RB}$.



Given: $\overline{RN} \cong \overline{GB}$, $\angle NRB \cong \angle GBR$
Prove: $RNBG$ is a parallelogram.

Commented [DB1]: <https://www.geogebra.org/calculator/myhffg3r>

Statement	Reason
$\overline{RN} \cong \overline{GB}$	Given
1.	2.
$\angle NRB \cong \angle GBR$	Given
$\triangle RGB \cong \triangle RNB$	3.
4.	If two lines are intersected by a transversal and the alternate interior angles are congruent, then the lines are parallel.
5.	6.
$\overline{RG} \parallel \overline{NB}$	If two lines are intersected by a transversal and the alternate interior angles are congruent, then the lines are parallel.
$ABCD$ is a parallelogram	7.

Use the statements and reasons provided to complete the two-column proof.

$\angle BRG \cong \angle RBN$	Corresponding Parts of Congruent Triangles are Congruent
$\angle RGB \cong \angle NBR$	Reflexive Property of Congruence
$\overline{RB} \cong \overline{RB}$	Transitive Property of Congruence
$\overline{RG} \cong \overline{NB}$	$SSS \cong$
$\overline{RN} \parallel \overline{GB}$	$SAS \cong$
$\overline{RG} \parallel \overline{NB}$	A parallelogram has two sets of opposite parallel sides.
	A parallelogram has two sets of opposite congruent sides.

Fill in the blanks with the correct terms to complete the statements.

8. A quadrilateral containing _____ set(s) of _____ sides is a parallelogram.
9. _____ interior angles of a parallelogram are _____.
10. The _____ of a parallelogram bisect each other.